



An efficient two-stage hybrid framework to evaluate vortex-induced vibration for bridge deck based on divergent vibration

Yubing Song, Zilong Ti^{*}, Yongle Li

Department of Bridge Engineering, Southwest Jiaotong University, Chengdu, 610031, China

ARTICLE INFO

Keywords:

Bridge deck
Vortex-induced vibration simulation
Divergent vibration
Long-span bridge
Aerodynamic damping
CFD

ABSTRACT

High-fidelity CFD (computational fluid dynamics) numerical modeling offers an attractive alternative of VIV investigation for bridge deck. Conventional strategy to perform VIV prediction using CFD is to directly characterize the two-way coupled free vibration of structure and flow which could become dramatically expensive in computational cost. To bridge the gap, this study presents an efficient two-stage hybrid approach that combines a divergent vibration simulation for fast aerodynamic damping extraction and a nonlinear model to predict the VIV. A negative structural damping is imposed in the motion equation to intentionally create a divergent vibration which significantly accelerates the growth of motion and takes less than 1/3 of the elapsed time compared with a conventional VIV simulation whereas maintains reasonable accuracy in aerodynamic identification. A Π -shaped bridge deck is employed as a demonstration of the proposed approach. The modeled steady-state VIV performance is well validated against wind tunnel tests. The effect of negative damping on instantaneous flow field, surface pressure and amplitude prediction are discussed and the optimal damping ratio achieving balanced accuracy and efficiency is suggested. The proposed framework shows favorable efficiency while simultaneously retains satisfactory accuracy and is a potential numerical alternative of VIV investigation for bridge deck.

1. Introduction

The vortex shedding generated by the separating boundary layer flow around a bluff body which imposes oscillating crosswise forces on the body, and the phenomenon of “lock-in” will occur when the vortex shedding frequency is close to the natural frequency of the body (Wu and Kareem, 2012). In the lock-in regions, oscillating crosswise forces lead to possible vortex-induced vibration (VIV), which is a nonlinear vibration with self-exciting and self-limiting characteristics. Long-span bridges are typical flexible structures with low eigenfrequencies and thus become vulnerable to VIV. Although the VIV on long-span bridge is not as dangerous as flutter with respect to structural safety, large amplitudes and consistent vibration would cause fatigue issues and endanger the safety of driving. Investigation of VIV performance is one of the most primary tasks for long-span bridges (Feng, 1968; Jeong and Hussain, 1995; Nariman, 2016; Naudascher and Wang, 1993).

Wind tunnel test is one of the most reliable methods to evaluate the VIV of bridge deck (Laima et al., 2018b; Laima and Li, 2015; Yang et al., 2015) whereas the drawbacks of high expenses on time and labor are

also repelling. The development of high-performance computation provides a favorable alternative to evaluate the aerodynamic performance of bridge decks using numerical technology, e.g., CFD (computational fluid dynamics) (Hallak et al., 2013; Nariman, 2016; Pan et al., 2007; Shimada and Ishihara, 1999). Bruno and Khris (2003) studied the prediction of two-dimensional numerical simulation for VIV on bridge deck, and tested the performance of various turbulence models including Reynolds average Navier-Stokes (RANS) model and large eddy simulation (LES) model. Ge et al. (2022) investigated the underlying mechanism of VIV on a large-span suspension bridge using CFD techniques. A wide range of gap-width ratios (0–15) for parallel box girders were selected to investigate the aerodynamic interactions using LES by He et al. (2021) which includes the flow structures, pressure distributions, mean values and standard deviations of the force coefficients. Laima et al. (2018a) used LES to investigate the Reynolds number sensitivity on flow which past a twin-box girder. Xu et al. (2017) used numerical fluid-structure interaction (FSI) approach to verify the VIV reduced-order model (ROM) technique. Liu et al. (2021) demonstrated the flow pattern development during VIV via the CFD technique. Pan

^{*} Corresponding author.

E-mail address: zilongti@swjtu.edu.cn (Z. Ti).

<https://doi.org/10.1016/j.jweia.2023.105316>

Received 15 October 2022; Received in revised form 11 January 2023; Accepted 11 January 2023

Available online 17 January 2023

0167-6105/© 2023 Elsevier Ltd. All rights reserved.

et al. (2007) used the three-dimensional LES turbulence model to study the mechanism of VIV mitigation on box girder by means of forced vibration and further verified the results via free vibration simulation. However, due to the relatively small aerodynamic damping and slow amplitude growth in free vibration, the computational cost to reach a steady-state response could be dramatically expensive. To reduce computational costs, some studies presented hybrid approaches in which the aerodynamic parameters or derivatives of bridge deck were simulated and identified, and the steady-state motions of VIV were predicted using semi-analytical model based on identified aerodynamic parameters. Iwamoto and Fujino (1995) proposed an extraction method to simultaneously extract aerodynamic derivatives of two-order modes by free vibration. Ge and Xiang (2008) presented the self-excited aerodynamic model associated with the numerical extraction method of aerodynamic derivatives. Gao et al. (2020) proposed an empirical model to describe aerodynamic nonlinearity during the so-called soft flutter of a bridge deck with a medium side ratio. Xu (2017) utilized numerical free vibration to extract the aerodynamic derivatives of the bridge deck and the weak coupling method was used to solve the fluid-structure interaction. Noguchi et al. (2020) extracted the aerodynamic damping and predicted the VIV amplitude using constant-amplitude forced vibration, which accurately predicted the “lock-in” wind speed region whereas the prediction of amplitude may be affected by the size of the spanwise domain and the Reynolds number, resulting in a large error between the results of predicted and test. Wang and Chen (2022) used an efficient high-order spectral element/Fourier method to simulate the aerodynamic damping and presented a varying-amplitude forced oscillation approach which allowed single CFD run to extract the aerodynamic damping. Some selective recent methods or frameworks for bridge VIV investigation are listed in Table 1. It indicates that there exist three mainstreams of VIV investigation for bridge deck. Numerical simulation for the fixed model is exclusively used to visualize the flow field and is not suitable to evaluate VIV amplitude (Sarwar and Ishihara, 2010). The conventional approach for amplitude prediction is to directly conduct a two-way coupled free vibration (Laima et al., 2013), but the computational costs could be extremely expensive and makes it unpractical. Another approach is to identify aerodynamic damping using forced vibration and predicts the VIV amplitude based on the modeled

aerodynamic parameters (Noguchi et al., 2020; Wang and Chen, 2022), and thus the computational costs are obviously reduced. However, the forced vibration must presume a steady-state amplitude which is usually acquired in advance, e.g., from model testing. Furthermore, VIV prediction often requires repeating simulations at multiple wind speeds and thus limits the forced vibration widely applied in engineering practice.

To bridge the gap, this study presents an efficient two-stage numerical framework to evaluate vortex-induced vibration and aerodynamic damping for bridge deck based on divergent vibration. The negative structural damping is imposed in the motion equation to intentionally create a divergent vibration which significantly accelerates the growth of motion. By using the time history of divergent vibration, the aerodynamic damping is identified via HT transformation and then is fed to a nonlinear VIV model to predict the steady-state amplitude. A Π -shaped bridge deck is employed as a prototype structure to demonstrate the approach. The proposed two-stage framework is delivered as follows: first, the dynamic behavior of the bridge deck subject to wind flow is modeled using a two-way coupled CFD simulation in which a negative structural damping ratio is imposed in the bridge deck's motion equation to intentionally create a divergent vibration and accelerate the growth of motion. Afterward, the amplitude-dependent aerodynamic parameters of vortex-induced force on the bridge deck is identified using the time history of structural motion and is fed to a nonlinear VIV model to predict the final VIV amplitude. The VIV performance associated with aerodynamic parameters evaluated using the two-stage framework are compared with wind tunnel tests. The effect of negative structural damping associated with the accuracy and efficiency of the proposed framework are also carefully discussed. The paper is organized as follows: the methodology as well as the procedure of the two-stage framework is described in Section 2. Section 3 shows the studying case of a Π -shaped bridge deck, and the experimental and numerical setups are introduced. Section 4 presents the results and discussions. The predicted VIV amplitudes are compared with wind tunnel tests. The effect of negative damping on instantaneous flow field, surface pressure and prediction accuracy are discussed and the optimal damping ratio meeting a balanced trade-off between accuracy and efficiency is suggested. Some conclusion remarks are drawn in Section 5.

2. Methodology

2.1. Numerical governing equations

Treating the bridge deck section as a rigid body, the flow around the bridge deck is modeled by means of the unsteady Reynolds-averaged Navier-Stokes (URANS) equations considering incompressible flow. According to Brusiani et al. (2013), the 2D URANS approach is equivalent to impose the correlation of the flow structures in the span-wise direction. Time-average the mass and momentum conservation equation to obtain the Reynolds-averaged equation of motion in the conserved according to Wilcox (1998).

$$\frac{\partial U_i}{\partial x_i} = 0 \quad (1a)$$

$$\rho \frac{\partial U_i}{\partial t} + \rho U_j \frac{\partial U_i}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} (2\mu S_{ij} - \rho \overline{u_i u_j}) \quad (1b)$$

where U_i is the mean velocity vector, x_i is the position vector, t is the time, ρ is the air density, u_i is the fluctuating velocity and the overbar represents the time average, P is the mean pressure, μ is the air viscosity, S_{ij} is the mean strain-rate tensor. The k - ω SST model is used to solve the closure problem of incompressible flows, which is integrated through the viscous sublayer without other terms. (Wilcox, 1998).

The numerical simulation in this study is performed in commercial software ANSYS Fluent 19.2. The governing equations are discretized by the finite volume method, and the control volumes are solved in the

Table 1
Summary of selective numerical frameworks for VIV investigation.

Classification	Literature	Approach/Model	Content
Fixed model	Ge et al. (2022)	LES	VIV and mitigation mechanism
	He et al. (2021)	LES	Flow structures and aerodynamic interference
	Zhang et al. (2020)	URANS	VIV and mitigation mechanism
Free vibration	Li et al. (2022)	URANS, Two-way couple	Characteristic of VIV “lock-in” regions
	Laima et al. (2013)	LES, Two-way couple	Investigation and mitigation of VIV
	Sarwar and Ishihara (2010)	LES, Two-way couple	VIV and mitigation mechanism
Forced vibration	Sarwar and Ishihara (2010)	LES, Forced vibration, Constant amplitude	VIV and mitigation mechanism
	Liu et al. (2021)	LES, Forced vibration, Constant amplitude	Flow pattern development during VIV
	Noguchi et al. (2020)	LES, Forced vibration, Constant amplitude	VIV amplitude prediction
	Wang and Chen (2022)	LES, Forced vibration, Constant & variable amplitude	VIV amplitude prediction, VIV mechanism

computational domain using the integral form of the conservation equations. The convection term adopts the central difference scheme, and the unsteady term adopts the second-order implicit scheme. SIMPLEC (SIMPLE-Consistent) algorithm for pressure-velocity decoupling associated with second-order upwind method is used.

2.2. Two-stage framework for VIV evaluation

The presented two-stage framework consists of two main steps. In the first step, the time-history motion of the bridge deck subject to uniform wind flow is modeled using the coupled CFD simulation in which a specific negative structural damping ratio is imposed in the bridge deck's motion equation to intentionally create a divergent vibration and accelerate the growth of motion. In the second step, the amplitude-dependent aerodynamic parameters of vortex-induced force on the bridge deck is identified from the time history of structural motion and then is fed to a nonlinear VIV model to predict the VIV amplitude. The detailed procedures will be introduced respectively.

2.2.1. Stage 1: divergent vibration simulation

Under the assumption of homogeneous condition in the span-wise direction, the VIV motion of bridge deck is usually analyzed as a two-dimension sectional model. The motion of a 2D rigid sectional model can be defined with two main components, which are the translation displacement in heaving direction and the angular displacement about the centroid of the section. This study aims to investigate the heaving VIV in vertical direction exclusively and therefore the torsional degree of freedom is reduced. The employed motion equation of bridge deck undergoing VIV can be expressed as:

$$m\ddot{h}(t) + 2\xi_{hs}\omega_{hs}m\dot{h}(t) + m\omega_{hs}^2h(t) = F_a \quad (2)$$

among which ξ_{hs} is the structural damping ratio of the bridge deck, ω_{hs} is the natural circular frequency of deck vertical displacement, and m is the mass per unit length of the bridge deck, h is the vertical displacement, and F_a is the simultaneous aerodynamic force on bridge deck.

Normally, the corresponding structural parameters in Eq. (2) including mass, frequency, damping ratio, etc., should be chosen as the exact values deployed in experimental tests or engineering practice to numerically reproduce the real vibration, which is the standard procedure to simulate the VIV in most studies. However, due to the high expense of CFD modeling and inherent slow growth of VIV phenomenon, the coupled simulation could become extremely time-consuming. In addition, the evaluation of VIV performance usually requires repeating the simulation at various wind speeds which dramatically increases the computational expense. According to the current consensus of wind engineering community, the vortex-induced force (VIF) on bridge deck can be well described by the nonlinear aerodynamic damping theory and once the core variable, i.e., aerodynamic parameter which is used to calculate the aerodynamic damping, is obtained, the VIV amplitude can be predicted accurately. The relevant aerodynamic parameters can be

identified from the time history of flexible kinds of vibration, e.g., decay-to-resonance, grow-to-resonance, etc.

Inspired by the statement above, to improve the efficiency, the structural damping ratio ξ_{hs} in Eq. (2) is intentionally set to a negative value, indicating an artificial negative-damping scenario, and the VIV vibration is expected to rapidly diverge rather than tends to reach a steady-state VIV due to the absence of structural damping. This manipulation mainly aims to accelerate the growth of motion and let the deck's displacement reaches a sufficient high amplitude within a short time. Two typical modeling time histories representing the growth of displacement using normal positive damping ratio $\xi_{hs} = 0.5\%$ and a negative damping ratio $\xi_{hs} = -0.3\%$ are illustrated in Fig. 1 (a) as a demonstration, and the technical details will be described in the following contents. It reveals that the displacement in the negative damping case diverges fast and rapidly grows to a relatively high amplitude whereas in the normal damping case, the displacement develops slowly, especially near the steady-state amplitude. It only takes about 7s for the negative damping case to reach the same amplitude of 0.8×10^{-3} m which the normal damping case finally reached after 40s. Moreover, by setting a larger absolute value of negative damping ratio will further benefit the growth of motion, as shown in Fig. 1 (b).

Divergent vibration is performed using the two-way coupled CFD simulation. The CFD solver is interacted with a User-Defined function (UDF) in which the structural divergent motion is simultaneously calculated. At each time step, the instantaneous aerodynamic force bridge deck is computed by the CFD solver and fed to the fourth-order Runge-Kutta algorithm to numerically solve the motion information including velocity and position of the bridge deck. Afterward, the motion information is returned to the CFD solver to update the grids and boundary condition for the next time step. To obtain a valid time history of divergent displacement, the deck is fixed at the first few cycles to let the flow field fully develop and is then released. Note that at the releasing time, say t_0 , the instantaneous aerodynamic lifting force is not zero and will cause an impulse excitation to the deck. Such an initial impulse acts like a Dirac delta function and could cause undesirable irregular motions in the time histories. For a positive-damping structure, such an initial disturbance will rapidly dissipate whereas for a structure with negative damping, the effect of this impulse on structural motion will consistently exist and causes inaccuracy in the following aerodynamic identification. Therefore, a modulation function is employed in this study to eliminate the effect of initial disturbance. The instantaneous aerodynamic force is modified by multiplying the following modulation function $T_m(t)$ as:

$$T_m(t) = \begin{cases} \frac{1}{2} \left[1 - \cos \pi \left(\frac{t-t_0}{T} \right) \right] & (t_0 \leq t \leq T) \\ 1 & (t > T) \end{cases} \quad (3)$$

where t is the simulation time, and T is the vibration period of the bridge deck. The modulation function allows the aerodynamic force following a smooth transition from zero to the actual amplitude within one period

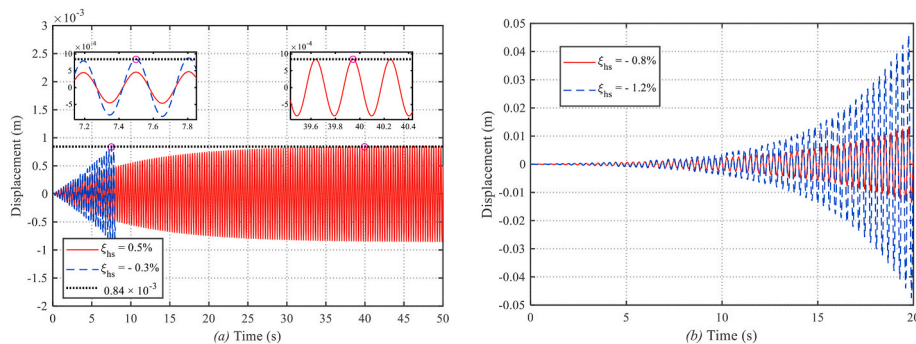


Fig. 1. Typical displacement time histories under various damping ratios.

and thus could effectively weaken the effect caused by the initial impulse excitation.

The divergent simulation can be stopped when the modeled divergent vibration reaches a certain amplitude and such an ending amplitude is quite customizable, depending on the maximal required amplitude in the aerodynamic damping parameters. One can flexibly choose the ending amplitude as long as the amplitude covers the potentially desirable VIV amplitude in engineering practice. Note that at lock-in wind speeds, the divergent vibration using negative damping is not necessarily always having a divergent trend. Since the aerodynamic damping of VIV has an inherent characteristic of amplitude-dependency, with the amplitude growing during the simulation, the aerodynamic damping will become positive from negative and may eventually cancel the negative structural damping, leading to a steady-state vibration of the deck.

2.2.2. Stage 2: aerodynamic damping and VIV prediction

The time history of divergent vibration in Fig. 1 contains the underlying aerodynamic damping information in VIF and could be reasonably identified and extracted. For a given time history signal of motion $y(t)$, the inherent time-varying amplitude and frequency can be decomposed using the Hilbert transform (HT) (Wang et al., 2020).

$$H[y(t)] = \tilde{y}(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{y(\tau)}{t - \tau} d\tau \quad (4a)$$

$$Y(t) = y(t) + i\tilde{y}(t) = A(t)\exp[i\theta(t)] \quad (4b)$$

$$A(t) = \sqrt{y^2(t) + \tilde{y}^2(t)} \quad (4c)$$

$$\theta(t) = \arctan \left[\frac{\tilde{y}(t)}{y(t)} \right] \quad (4d)$$

$$\omega(t) = 2\pi f(t) = \frac{d\theta(t)}{dt} \quad (4e)$$

where $A(t)$ is the time-varying amplitude, $\omega(t)$ is the time-varying frequency. For structures experiencing VIV, the time-varying frequency is approximately the fundamental frequency. Fitting the discrete time-varying amplitude signal obtained by HT transformation to a continuous function to accurately express the correlation between amplitude and aerodynamic damping and facilitate the calculation of its derivative with respect to time. The displacement of VIV usually follows the exponential growth pattern due to the inherent damping characteristics of VIF, and therefore the trend of amplitude variation with respect to time well conforms to the following exponential model, where a_0 , a_1 , t_0 , and b are fitting parameters:

$$A(t) = a_0 \exp \left[- \left(\frac{t - t_0}{a_1} \right)^2 \right] + b \quad (5)$$

Note that the fitting procedure above is an alternative approach. One can use the discretized signal $A(t)$ to numerically proceed the analysis and not necessarily conduct the curve fitting. However, using the fitting curve shows certain advantages on smoothing numerical noises and analytical derivations. Given the curve of $A(t)$, the time-varying damping ratio $\xi_h(t)$ of the coupled system can be estimated by the following formula:

$$\xi_h(t) = - \frac{1}{\omega_h} \frac{d \ln[A(t)]}{dt} \quad (6)$$

Note that the damping ratio above is the total damping ratio which is the summation of the structural damping ratio and aerodynamic damping ratio.

For bridge deck free experiencing VIV, the vertical self-excited lift ($L_{se}(t)$) can be expressed in terms of the aerodynamic derivatives

(Scanlan, 1978; Scanlan and Tomko, 1971)

$$L_{se}(t) = \frac{1}{2} \rho U^2 (2b) \left(kH_1^* \frac{\dot{h}}{U} + k^2 H_4^* \frac{h}{b} \right) \quad (7)$$

where $b = B/2$ is the half width of bridge deck, $k = \omega b/U$ is the reduced frequency, and ω is the vibration circular frequency, H_1^* and H_4^* represent the aerodynamic parameters related to the aerodynamic damping and the aerodynamic stiffness. At the vicinity of the vortex lock-in region, H_1^* and H_4^* are functions of the dimensionless vibration amplitude $A = A_h/D$ (A_h is the amplitude of VIV) and the reduced frequency k , which is $H_1^* = H_1^*(k, A)$, $H_4^* = H_4^*(k, A)$ (Chen, 2013). It indicates that VIV is amplitude-dependency. When the VIV occurs, the vibration of the bridge deck can be described as:

$$m \left(\ddot{h} + 2\xi_{hs}\omega_{hs}\dot{h} + \omega_{hs}^2 h \right) = \frac{1}{2} \rho U^2 (2b) \left(kH_1^* \frac{\dot{h}}{U} + k^2 H_4^* \frac{h}{b} \right) \quad (8)$$

Within lock-in region, the change in the total stiffness of the structure caused by H_4^* is usually small and can be reasonably neglected (Wang and Chen, 2022) and thus Eq. (8) can be described as:

$$\ddot{h} + 2\xi_h\omega_{hs}\dot{h} + \omega_{hs}^2 h = 0 \quad (9)$$

$$\xi_h = \xi_{hs} + \xi_{ha} = \xi_{hs} - \frac{1}{2} \left(\frac{\rho b^2}{m} \right) H_1^* \quad (10)$$

where ξ_{ha} is the aerodynamic damping ratio. VIV occurs when the total damping of the system meets the criteria $\xi_h = \xi_{hs} + \xi_{ha} = 0$. At a certain wind speed, given a structural damping ξ_{hs} of testing model or prototype bridge, one can find the point $\xi_{ha} = -\xi_{hs}$ from the aerodynamic damping curve. Since the aerodynamic damping curve is the function of amplitude which is shown in Fig. 2 (a), where the abscissa axis represents the reduced amplitude of the structure (D is the height of the deck) and the vertical axis indicates the aerodynamic damping ratio, the final steady-state amplitude of VIV can be easily predicted. If the desirable condition $\xi_{ha} = -\xi_{hs}$ is beneath the lower limit of aerodynamic curve, it suggests that the specific wind speed is out of lock-in region and the VIV will not occur. If the desirable point $\xi_{ha} = -\xi_{hs}$ is beyond the upper limit of the aerodynamic curve, it probably suggests that the ending amplitude is not sufficiently large enough and one may need to continue the simulation to obtain more motion data at a higher amplitude.

The standard procedure above describes how to predict the VIV amplitude from divergent vibration. However, at some wind speeds, one may observe a steady irregular vibration with limited amplitude in the simulation rather than a divergent one, as displayed in Fig. 2 (b). This phenomenon suggests that even given a negative structural damping, the total damping of the system is still large enough to ensure a steady vibration, and therefore the aerodynamic damping is positive at the wind speed. It is safe to say that the VIV will not occur at these specific wind speeds, i.e., out of lock-in region. To sum up, the negative structural damping is intentionally added into the motion equation and integrated with the two-way coupled CFD simulation. During the simulation, the modulation function is deployed to realize a smooth transition of aerodynamic force from zero to actual amplitude within one cycle and weaken the irregular motions in time history caused by the initial impulse excitation. Using the time history of divergent vibration, the aerodynamic damping is identified via HT transformation and then is fed to a nonlinear VIV model to predict the steady-state amplitude, the proposed framework and relevant procedures are portrayed as a flow-chart in Fig. 3.

3. Numerical and experimental setup

3.1. Experimental setup

A Π -shaped deck utilized in a 608m long-span cable-stayed bridge is

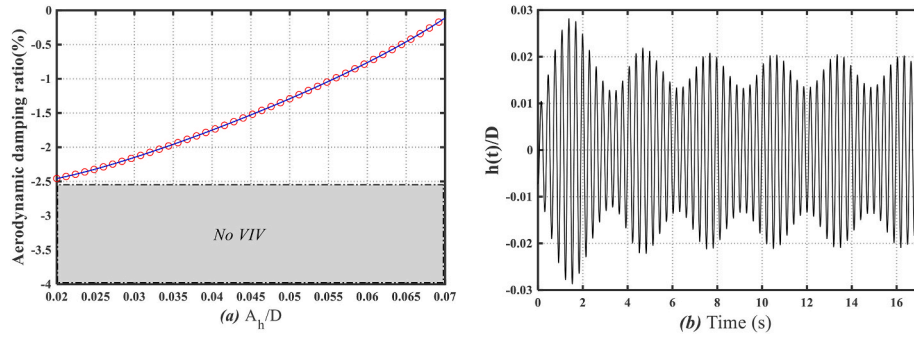


Fig. 2. Typical scenarios for VIV prediction: (a) modeled aerodynamic damping curve and VIV amplitude; (b) time history of steady vibration.

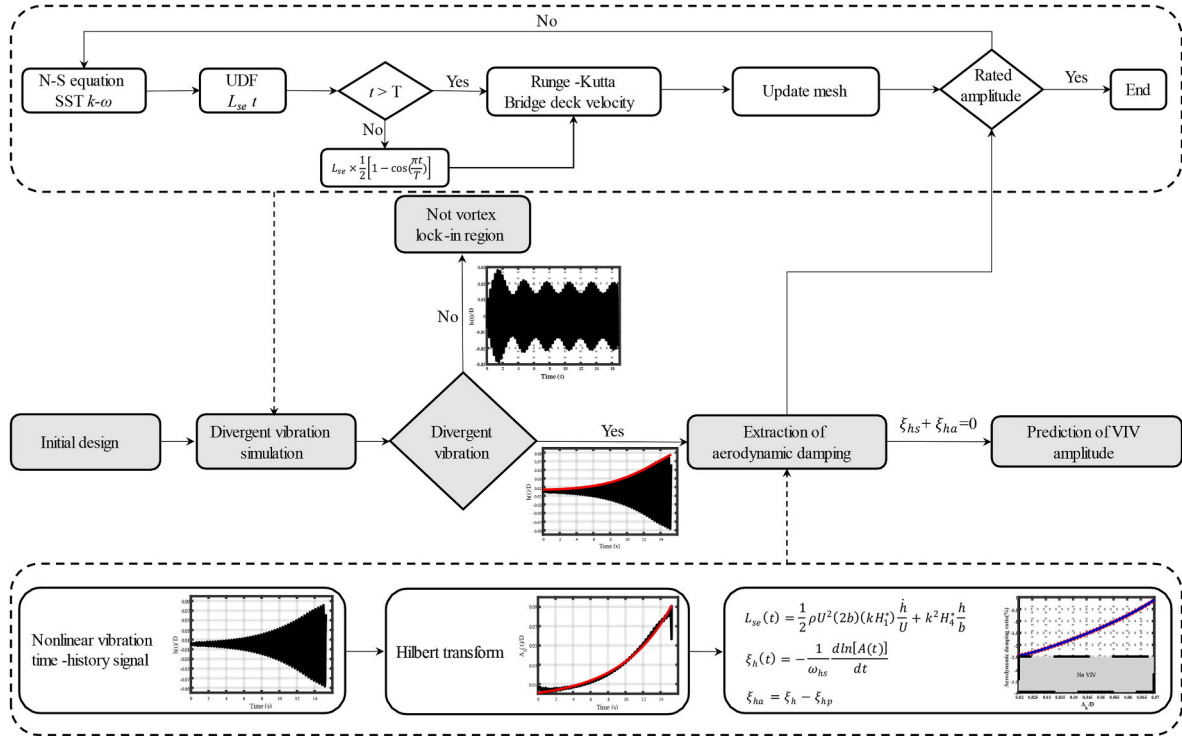


Fig. 3. Flowchart of the proposed two-stage framework for VIV amplitude evaluation.

studied. The testing model, as shown in Fig. 4, has a scale ratio of 1:60. The dimension of the testing model is 2.10 m, 0.60 m and 0.06 m in length L , width B and height D , respectively, and the mass per unit length of the bridge deck is 13.87 kg/m. The vertical frequency of the testing model is 3.27Hz. The vertical damping ratio (ξ_{hs}) measured on site is 0.65%.

The sectional wind tunnel test of the scaled bridge deck model is carried out in the XNJD-1 wind tunnel. The cross-section of the wind tunnel is 2.0m and 2.4m in height and width, and allows the wind speed to flexibly change from 0 m/s to 30.0 m/s. The testing model is made of

ABS high-strength plastic and pine wood to ensure a sufficient rigidity in span-wise direction. The model is elastically suspended by four pairs of linear springs to evaluate its VIV steady-state amplitude, as displayed in Figs. 5 and 6.

3.2. Numerical setup

The employed numerical model is exhibited in Fig. 7. The dimension of the computational domain is $20B$ in length and $30D$ in height and is divided into three sub-domains. The external rigid area is discretized

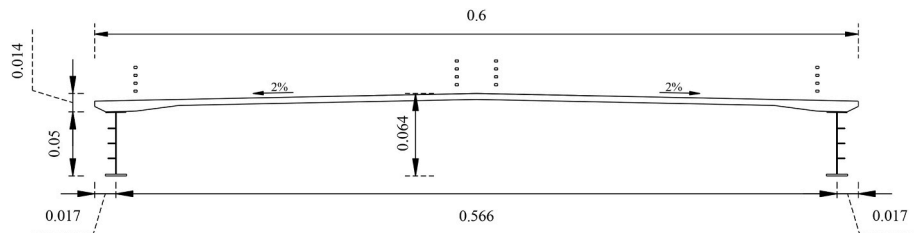


Fig. 4. Deck configuration (unit: m).

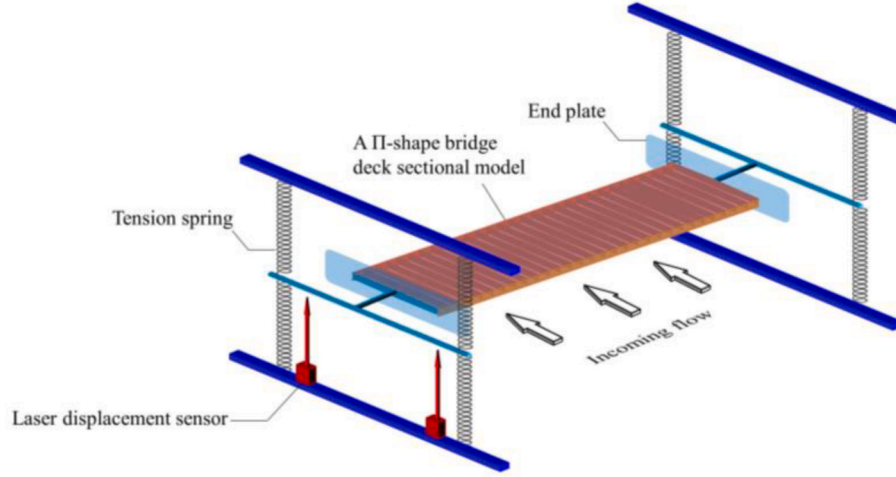


Fig. 5. Experimental configuration of model test.



Fig. 6. Testing model of bridge deck.

using structured quadrilateral grids. For the external rigid area far from the deck, radial grids which gradually increases the grids size with the distance from the bridge deck are used to achieve a balance between numerical efficiency and accuracy. The unstructured quadrilateral grids are used to discretize the computational area near the deck, which is more geometrically flexible than the structural grids and it is easier to transition from the boundary layer to the outer grid. In order to ensure the quality of the near-deck grids during the fluid-structure interaction calculation (Tamura et al., 1998), the area near the deck is defined as an internal rigid domain that the rigid domain grids move with the deck to ensure a high accuracy simulation of the flow field near the deck. The deformed area is discretized by triangular grids. The bridge deck is set as the wall boundary. The grid details are shown in Fig. 8 and the total number of grids is about 300,000. In this study, the time increment is set $\Delta t = 5.0 \times 10^{-4}$ and $y^+ < 1$ is satisfied for the first layer grids in most cases (Ferziger et al., 2002). With respect to the structural parameters, to

perform the divergent vibration simulation, the default negative damping ratio is set to -0.3% . The structural motion is resolved simultaneously using the aforementioned UDF. Other parameters, e.g., mass, stiffness, are employed as the experimental values shown in Table 2.

Convergence analysis is conducted regarding the influence of grid number and time step of the simulated data on the target indicators, i.e., the aerodynamic coefficient ($C_D = 2F_D/\rho U^2 D$; $C_L = 2F_L/\rho U^2 B$; $C_M = 2M_T/\rho U^2 B^2$), where F_D , F_L and M_T are drag, lift and moment of the deck respectively. In addition, Strouhal number ($S_t = f_v D/U$) is also an important indicator to be verified, where f_v is the frequency of vortex shedding. The results for aerodynamic coefficient of deck are shown in Table 3, comparing the results of five strategies, it can be seen that the simulation results of the original scheme (case 3) are accurate enough.

4. Results and discussion

4.1. Aerodynamic damping and steady-state VIV amplitude

The VIV performance of the testing deck model is evaluated using the proposed two-staged approach and the results including VIV amplitude and aerodynamic parameters are compared with experimental data. The predicted VIV amplitude of the deck against various reduced wind speeds U/fD using the proposed framework are presented in Fig. 9 and validated with the wind tunnel testing results. With respect to the testing results, one can see that there exist two lock-in regions in which the primary lock-in region at $U/fD = 12.56 - 25.02$ shows a large width of wind speeds and prominent amplitudes whereas the secondary region concentrates at $U/fD = 7.91 - 9.47$ with much smaller amplitudes. Note that unexpected torsional VIV occurs in the testing at $U/fD = 20.21$ as marked in light grey, which makes the motions of testing model become the combination of vertical and torsional vibration. Regarding the predicted results, it shows that the predicted VIV amplitudes have favorable agreement with the testing results in terms of both magnitude and wind

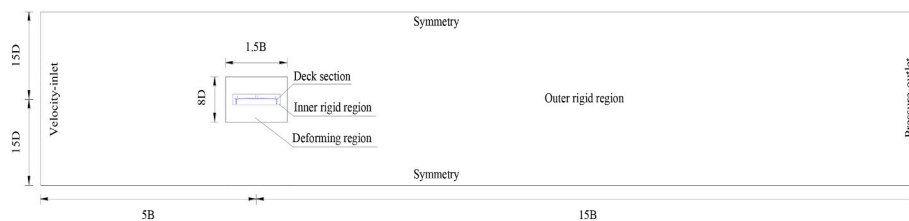


Fig. 7. Computational domain and boundary conditions.

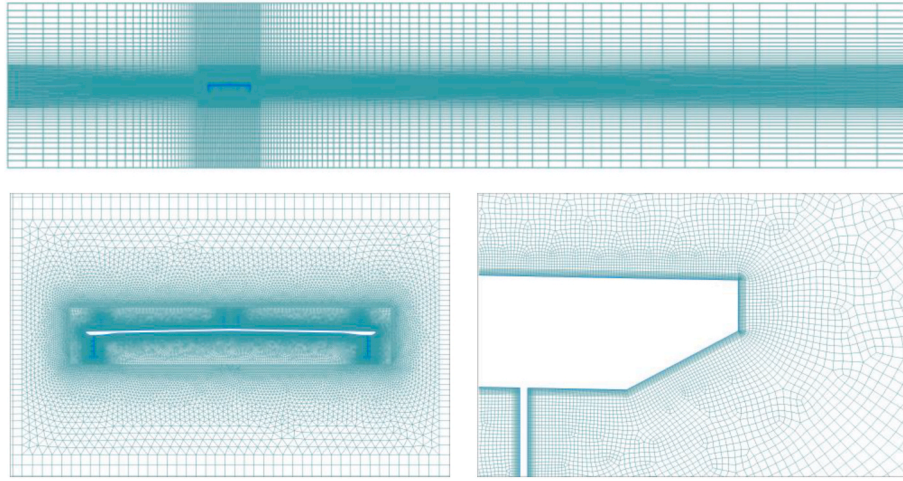


Fig. 8. Grid details of employed mesh.

Table 2
Parameters of wind tunnel test.

Property		Unit	Prototype	1:60 Model
Length	L	m	/	2.10
Width	B	m	36.00	0.60
Height	D	m	3.85	0.06
Mass per unit length	m	kg/m	4.99×10^4	13.87
Vertical frequency	f_{vs}	Hz	0.31	3.27
Vertical damping ratio	ξ_{hs}	%	/	0.65

Table 3
Convergence analysis of simulation strategy.

Case ID	Grid number	Time step	C_D	C_L	C_M	S_t
1	30×10^4	0.0001	1.17	-0.21	0.03	0.07
2	30×10^4	0.0003	1.16	-0.19	0.03	0.07
3	30×10^4	0.0005	1.17	-0.21	0.03	0.07
4	35×10^4	0.0005	1.19	-0.20	0.03	0.07
5	40×10^4	0.0005	1.18	-0.18	0.03	0.07

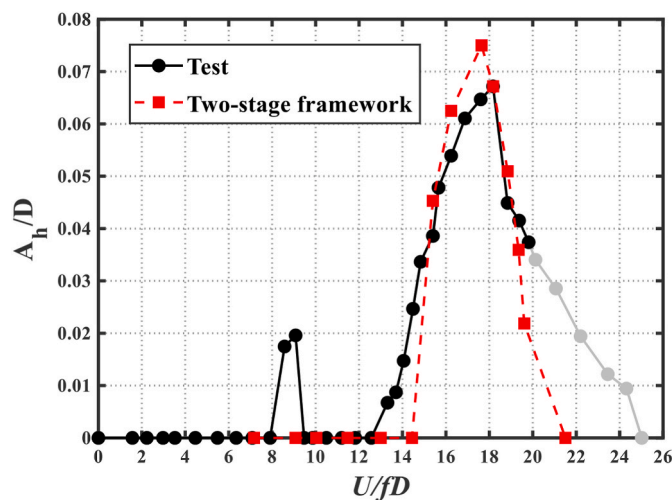


Fig. 9. Predicted and experimental VIV amplitudes of testing model.

speeds. However, some deviations between predicted and testing results are still observed and the secondary “lock-in” region within $U/fD = 7.91 - 9.47$ is absent in the predicted results. According to the study of

two “lock-in” regions characteristic of VIV of Π -shaped deck by Li et al. (2022), the flow field characteristics of two “lock-in” region are obviously different and the secondary VIV region is related to the “Karman vortex street” at the trailing edge of the section (Li et al., 2022; Wang and Chen, 2022). The intense lift oscillation leads to a poor capture ability of the deck vibration to the vortex shedding frequency. Therefore, the vibration of the deck is more likely to be dominated by a forced component rather than the self-exciting component. Since the aerodynamic model is appropriately simplified to a fully self-excited model in this framework, it is failure to predict the VIV in the secondary lock-in region. Another potential reason causing this issue is the relatively simple numerical scheme and turbulent model employed in this study, i. e., a 2 dimensional RANS model. More complicated turbulence structures around bluff body may require a more advanced scheme to reproduce, such as 3-dimensional LES model. Concerning the primary lock-in region, the proposed framework gives overall satisfactory prediction but slightly overestimates the maximum amplitude at $U/fD = 17.64$. In addition, the proposed framework tends to underestimate the width of lock-in region with the predicted starting wind speed of $U/fD = 14.44$ compared with testing results $U/fD = 12.57$. At the starting wind speed of lock-in region, the VIV is very sensitive to initial disturbance and damping ratio, and the numerical simulation may fail to accurately reproduce the aerodynamic force as well as the flow around the deck. Beyond reduced wind speed $U/fD = 20.12$, due to the unexpected coupled vertical-torsional VIV experienced in the testing model, the comparison is unnecessary since the torsional motion is absent in the simulation and out of the scope in this study. Nevertheless, the predicted results show reasonable pattern at the main lock-in region $U/fD = 14.44 - 19.81$ which well agrees with the testing results.

To further investigate the modeling results, the time history of dimensionless displacement $h(t)/D$ at four typical reduced wind speeds $U/fD = 7.17, 14.44, 17.64, 18.84$ using the proposed divergent vibration approach are demonstrated in Fig. 10. For results at $U/fD = 7.17, 14.44$ in Fig. 10 (a) and (b), the time histories of displacement show stable ‘beat’ phenomenon rather than a divergent vibration. It suggests that even given a negative damping, the deck at these wind speeds is aerodynamically stable and the input motivating energies from vortex-shedding are smaller than the dissipating energies caused by aerodynamic damping forces, and therefore VIV will not occur at these wind speeds, i.e., beyond “lock-in” regions. At reduced wind speeds $U/fD = 17.64$ and 18.84 , the displacements are experiencing the expected divergent vibration and the aerodynamic parameters can be evaluated following the proposed two-stage framework, as shown in Fig. 10 (c) and (d). The fitting curves of dimensionless amplitude A_h/D marked in red show that the fitting model can well describe the growth of amplitude

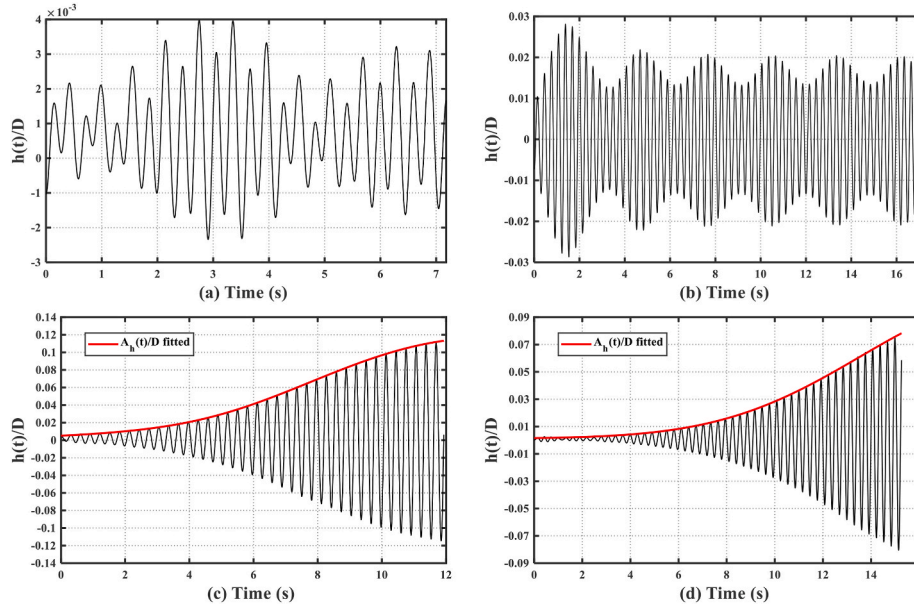


Fig. 10. Time history samples of deck displacement $h(t)/D$ and partial fitting results: (a) $\xi_{hs} = -0.3\%$, $U/fD = 7.17$; (b) $\xi_{hs} = -0.3\%$, $U/fD = 14.44$; (c) $\xi_{hs} = -0.3\%$, $U/fD = 17.64$; (d) $\xi_{hs} = -0.3\%$, $U/fD = 18.84$

against time. The identified aerodynamic parameters H_1^* using the modeled divergent vibration time histories at $U/fD = 15.40$, 16.24 , 17.64 , 18.17 , 18.84 , 19.37 are shown in Fig. 11. It reveals that the aerodynamic parameters representing the aerodynamic damping show obvious amplitude-dependency which is well known as one of the most significant inherent characteristics in bridge deck VIV.

4.2. Effect of negative damping

The negative damping which is employed to control the growth of divergent vibration plays a main role in the proposed framework. A relatively larger absolute value of negative damping ratio will significantly accelerate the divergent vibration and shorten the computational time to reach a desirable amplitude whereas the drawback is also obvious that the amplitude resolution in displacement signals will drop, which may affect the final prediction accuracy. To investigate the effect of negative damping ratio on the performance of proposed framework,

four negative damping ratios, i.e., $\xi_{hs} = -0.3\%$, -0.6% , -1% , -1.5% are numerically tested, the time-history of displacement and aerodynamic damping curves at $U/fD = 17.64$ are shown in Fig. 12. The resultant steady-state VIV amplitudes are demonstrated in Fig. 13. It reveals that the selection of negative damping has certain influence on accuracy and divergent speed. With the increase of the absolute value of the negative damping ratio, the predicted amplitude of VIV tends to increase. The predicted VIV amplitude of $\xi_{hs} = -0.3\%$ has the best agreement with experimental amplitude whereas when $\xi_{hs} = -1.5\%$, the resultant amplitudes deviate from experimental results the most. In general, all the tested negative damping ratios can give reasonable VIV amplitudes whereas with the increasing of negative damping ratios, the errors in the resultant VIV amplitudes tend to increase.

In order to have more in-depth investigation on the reasons causing the errors, the instantaneous flow fields of $U/fD = 17.64$ around the deck at the approximately same displacement amplitude in different damping ratio cases are characterized and displayed in Fig. 14. The

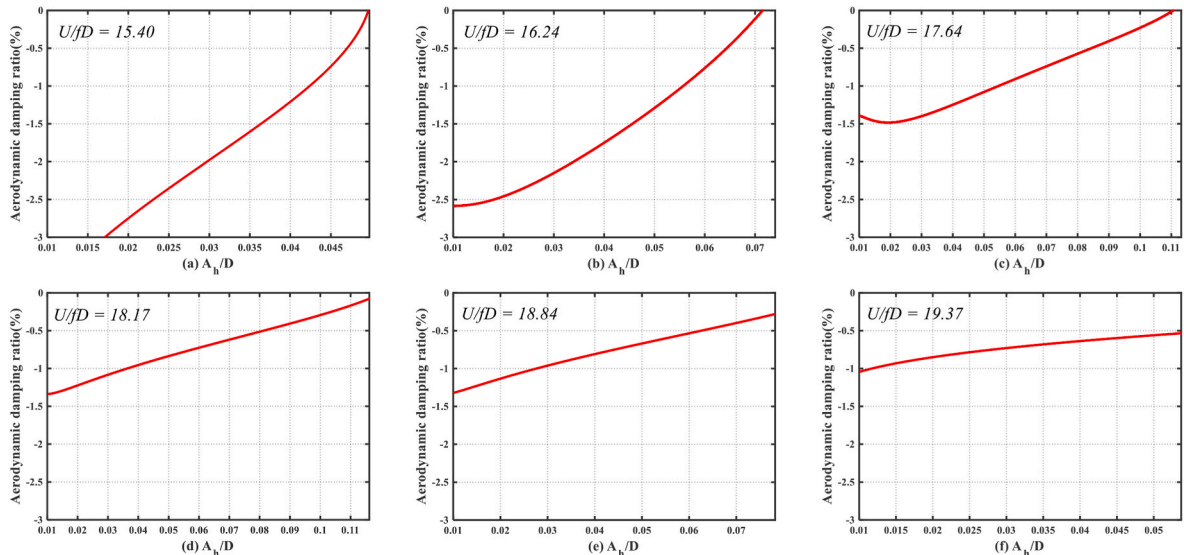


Fig. 11. Results of modeled aerodynamic damping curves.

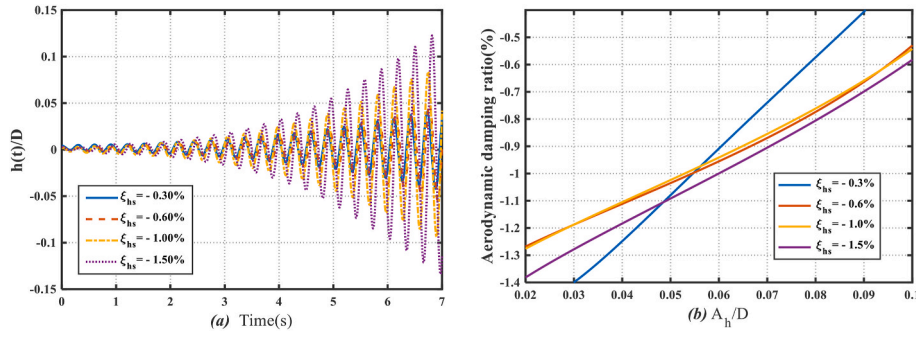


Fig. 12. Divergent vibration using various damping ratio: (a) Time-history of displacement; (b) identified aerodynamic damping curves.

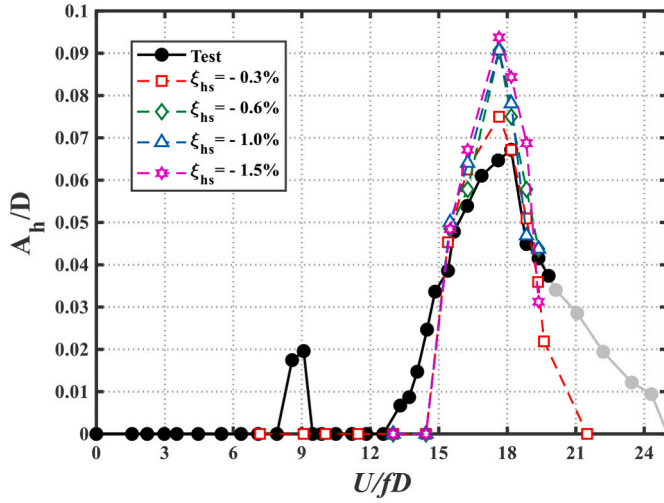


Fig. 13. Predicated VIV amplitudes with different negative damping ($\xi_{hs} = -0.3\%$, -0.6% , -1.0% , -1.5%).

conventional coupled VIV simulation in which the structural damping ratio is set to normal, i.e., the value of $\xi_{hs} = 0.65\%$ in the wind tunnel test is also carried out for comparison. Note that the instantaneous amplitude is chosen as the steady-state one in normal structural damping case of $A_h/D = 0.082$. Given the same amplitude, the flow fields are expected to share the similar pattern. It can be seen in Fig. 14 that the resultant flow fields using different damping ratio are highly similar. The shape, scale, and locations of vortices around decks remain consistent with the steady state normal structural damping case, which

provides reasonable proofs to the validity of the proposed framework from the perspective of flow field. The surface pressure coefficients ($C_p = p/(0.5\rho U^2)$, where p is the instantaneous pressure) on the bridge deck are characterized for further investigation, as shown in Fig. 15. The surface pressure coefficients C_p obtained under different damping ratio cases yield a similar pattern at most of the positions while some obvious differences at $0.1m$ on the lower surface and near the edges, i.e., -0.2 to $-0.3m$, $0.2m$ on the upper surface are observed. These differences in pressure will cause errors in instantaneous aerodynamic forces on the deck and the growing rate of the displacement time histories, and eventually leads to errors in the damping identification and VIV amplitude prediction. One reasonable explanation for this phenomenon is that during the divergent vibration, the flow field is essentially unstable due to the consistently growing of deck's motion rather than a steady-state one in real VIV and this instability of flow field can cause discrepancies in the instantaneous pressure and aerodynamic forces on the deck. The instability of flow field grows with the negative damping ratios and as a result, the VIV amplitude predictions are affected. Using the divergent vibration for VIV prediction assumes that the amplitude-varying aerodynamic damping is approximately equivalent to the steady-state one, of which assumption is also extensively used in aerodynamic parameter identification processes including decay-to-resonance and growth-to-resonance methods. However, due to the inherent complexity of wind flow, they are not essentially identical, and the effect becomes greater when the absolute value of negative damping ratio is increasing.

4.3. Trade-off between efficiency and accuracy

The results presented above reveal that prediction accuracy seems to slightly decrease with the increasing absolute value of negative damp-

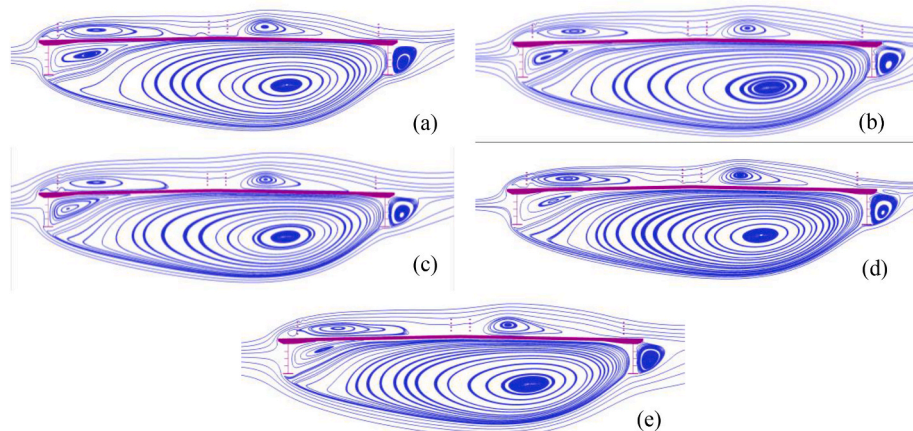


Fig. 14. Numerical streamlines around deck around the same maximum amplitude in different damping ratio cases: (a) $\xi_{hs} = 0.65\%$; (b) $\xi_{hs} = -0.30\%$; (c) $\xi_{hs} = -0.60\%$; (d) $\xi_{hs} = -1.00\%$; (e) $\xi_{hs} = -1.50\%$.

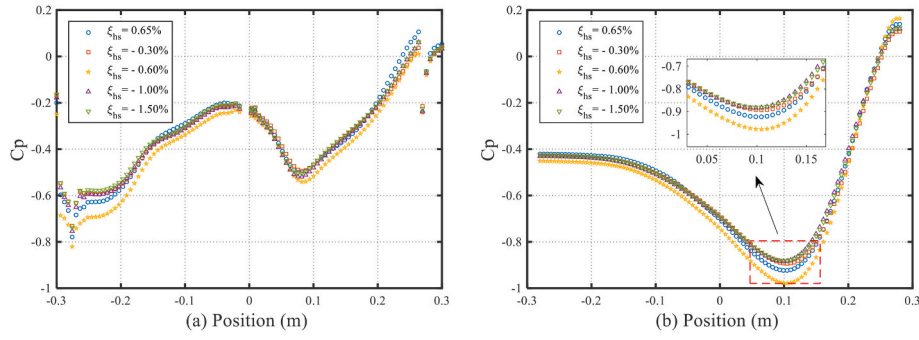


Fig. 15. Instantaneous pressure coefficient on upper and lower surfaces for different damping ratio ($\xi_{hs} = 0.65\%, -0.30\%, -0.60\%, -1.00\%, -1.5\%$; $U/fD = 17.64$): (a) The pressure coefficient of upper deck; (b) The pressure coefficient of upper deck.

ing. However, on the other hand, by setting a larger absolute value of negative damping ratio, despite the slight drop in accuracy, the computational efficiency could be dramatically improved. The computational time using different negative damping ratios is investigated following the proposed framework and compared with the conventional coupled VIV simulation in which the structural damping ratio is set to normal, i.e., the experimental value of $\xi_{hs} = 0.65\%$. The computational time is defined as the elapsed CPU time to reach the steady-state amplitude in the conventional case. All the simulation in this study is performed on an Intel(R) Core i9-12900 @2.40 GHz computer. The results at a typical wind speed $U/fD = 18.17$ are analyzed as an example in Fig. 16. It indicates that the conventional method using normal damping ratio $\xi_{hs} = 0.65\%$ takes 88 cycles to reach the steady-state amplitude whereas the negative damping cases of -0.3% to -1.5% take only 32–18 cycles to reach the same amplitude. With respect to the computational time, the conventional case takes up to 102 h to achieve a steady-state vibration whereas the negative damping cases of -0.3% to -1.5% dramatically reduce the computational time to 36.8–20.7 h. It often requires multiple simulations to evaluate the VIV performance at all possible encountering wind speeds, the proposed framework could save considerable time for VIV investigation of bridge deck.

A proper selection of negative damping ratio is essentially to make a trade-off between accuracy and efficiency. To roughly evaluate the comprehensive performance, a trade-off factor is defined for each damping ratio as follow:

$$R = \frac{1}{N} \sum_{i=1}^N \left| \frac{(A_h/D)_{p,i} - (A_h/D)_{test,i}}{(A_h/D)_{test,i}} \right| \times \frac{1}{N} \sum_{i=1}^N \frac{T_{c,i}}{T_{c,0}} \quad (11)$$

where $(A_h/D)_{p,i}$ and $(A_h/D)_{test,i}$ are the dimensionless VIV amplitudes from simulation and wind tunnel model test at the i_{th} wind speed. N is the number of investigated wind speeds. T_c is the computational time and $T_{c,0}$ is the computational time of the conventional VIV simulation method using normal damping ratio. The first term in Eq. (10) represents the averaged relative error of predicted VIV amplitudes against

experimental results and the second term represents the averaged relative time consumed compared with the conventional approach. R decreases with both computational time and relative error. Such a variable can be employed to roughly evaluate the comprehensive performance of a certain negative damping ratio considering both efficiency and accuracy, and the negative damping ratio achieves the minimal R can be reasonably considered as the potentially optimal damping ratio. Nine negative damping ratios varying from -0.3% to -1.5% under 3 wind speeds U/fD of 18.17, 18.84 and 19.37 within the lock-in region are investigated. The averaged computational time and relative error of 30 investigated cases are listed in Table 4, and the resultant trade-off factors R are displayed in Fig. 17. It reveals that damping ratio of -0.3% seems to yield the minimal R , i.e., 2.5 among all the investigated damping ratios whereas the damping ratio of -0.8% leads to the maximal R , indicating a relatively worse performance. Though the computational time rapidly decreases with the absolute value of damping ratio, but due to the increasing errors, R tends to increase. As a special case of normal damping ratio 0.65%, it can be seen that the trade-off factor R is up to 8.3 which is around 3.3 times greater than the damping ratio of -0.3% , revealing that compared with the conventional numerical simulation, the proposed framework shows superiority and could achieve favorable efficiency while simultaneously remains satisfactory accuracy.

The main advantage of the proposed framework includes but not limited to favorable efficiency and accuracy compared with other simulation approaches to evaluate VIV for bridge deck. Besides the time-consuming conventional approach, there exists some other methods via forced vibration of constant amplitude or forced vibration of time-varying amplitude (Noguchi et al., 2020; Wang and Chen, 2022). The forced vibration of constant amplitude must presume a steady-state amplitude of which is usually obtained beforehand, e.g., by means of model testing and therefore it is more of a hindcast method than a prediction method. Regarding the forced vibration of time-varying amplitude, in which the deck is forced to oscillate following a presumed growing function of amplitude and the instantaneous aerodynamic force is used for VIV prediction. However, due to the uncertain

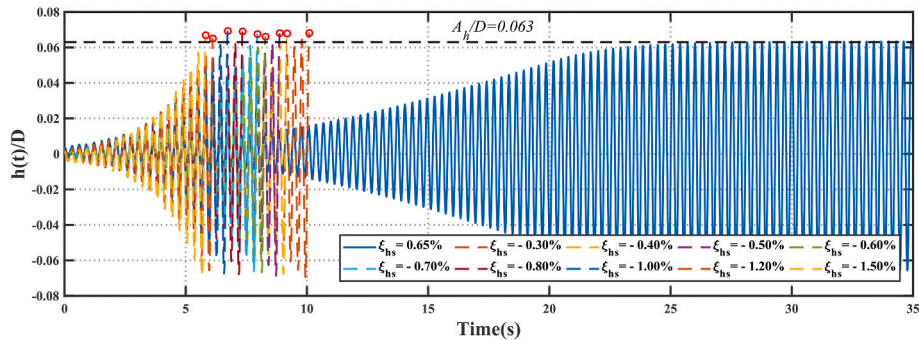
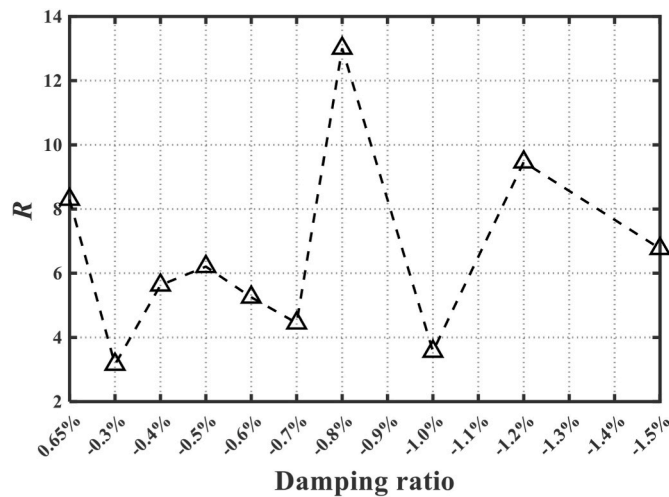


Fig. 16. Displacement time history using various damping cases at $U/fD = 18.17$

Table 4

Averaged computational time and relative error in different damping ratio cases.

Case ID	U/fD	ξ_{hs} (%)	Averaged elapsed CPU time (h)	Averaged relative error (%)
1	18.17	0.65	116.3	8.3
2	18.84	0.65		
3	19.37	0.65		
4	18.17	-0.30	41.03	9.0
5	18.84	-0.30		
6	19.37	-0.30		
7	18.17	-0.40	39.9	16.4
8	18.84	-0.40		
9	19.37	-0.40		
10	18.17	-0.50	37.2	19.4
11	18.84	-0.50		
12	19.37	-0.50		
13	18.17	-0.60	35.7	17.1
14	18.84	-0.60		
15	19.37	-0.60		
16	18.17	-0.70	33.8	15.3
17	18.84	-0.70		
18	19.37	-0.70		
19	18.17	-0.80	32.6	46.4
20	18.84	-0.80		
21	19.37	-0.80		
22	18.17	-1.00	29.2	14.2
23	18.84	-1.00		
24	19.37	-1.00		
25	18.17	-1.20	25.3	43.5
26	18.84	-1.20		
27	19.37	-1.20		
28	18.17	-1.50	21.5	36.6
29	18.84	-1.50		
30	19.37	-1.50		

**Fig. 17.** Tradeoff factor R of different damping ratios.

aerodynamic performance of the to-be-evaluated deck, the growing rate is hard to determine and either a too large or too small will cause inaccuracy or additional computational time in VIV prediction. In addition, it is reported that the numerical aerodynamic forces seem to become unstable with forced varying-amplitude (Wang and Chen, 2022). The proposed approach based on divergent vibration by means of negative structural damping is essentially equivalent to giving the deck a velocity-dependent excitation and therefore such an excitation is auto-adjustive. If the aerodynamic force is relatively high, the resultant growing rate will become greater whereas if the aerodynamic force is limited, the growing rate will drop as well. Such an advantage will make the resolution in the amplitude auto-adjustive, and eventually benefit the aerodynamic damping identification and VIV prediction.

5. Conclusion

In this study, a two-stage numerical framework based on divergent vibration is presented to evaluate the VIV performance of bridge deck. A negative structural damping is intentionally added into the motion equation and integrated with the two-way coupled CFD simulation. A modulation function is deployed to realize a smooth transition of aerodynamic force from zero to actual amplitude within one cycle and weaken the irregular motions in time history caused by the initial impulse excitation. Using the time history of divergent vibration, the aerodynamic damping is identified via HT transformation and then is fed to a nonlinear VIV model to predict the steady-state amplitude. The predicted VIV amplitudes for a H-shaped bridge deck are well validated against wind tunnel tests. The effect of negative damping on computational accuracy is discussed and the optimal damping ratio achieving a balanced trade-off between accuracy and efficiency is suggested.

Validation against wind tunnel test reveals that the proposed framework can successfully predict the main lock-in region associated with VIV amplitudes within $U/fD = 12.56 - 25.02$ and yields favorable agreement with experimental results. However, the presented results fail to predict the secondary lock-region of $U/fD = 7.91 - 9.47$ with insignificant amplitudes. The selection of negative damping ratio has certain impacts on the final VIV amplitude prediction. With the increasing of absolute value of negative damping ratio from -0.3% to -1.5% , the accuracy in VIV prediction tends to slightly drop. This phenomenon is due to the instability of flow field during the divergent vibration and the reasons have been discussed and explained from the perspective of instantaneous flow field and aerodynamic pressure. However, by utilizing a larger absolute value of negative damping ratio, despite the slight drop in accuracy, the computational efficiency could be dramatically improved. The conventional simulation approach takes up to 100+ hours to achieve a steady-state vibration whereas the proposed framework using damping ratio of -0.3% to -1.5% dramatically reduces the computational time to 36.8–20.7 h. To gain a balanced trade-off between computational accuracy and efficiency, 30 numerical cases using various damping ratios at different lock-in wind speeds are tested to evaluate the prediction performance based on a trade-off factor which comprehensively considers the errors and elapsed CPU time. The results suggest that the damping ratio of -0.3% seems achieve an optimal trade-off that significantly reduces the computational time from 116.3 h to 41.03 h while simultaneously maintains a favorable accuracy of 7.2%. In addition, the proposed approach based on divergent vibration is equivalent to giving the deck a velocity-dependent excitation and therefore such an excitation is auto-adjustive. Compared with other hybrid VIV simulation approaches, e.g., forced vibration with constant/varying amplitude, the growing amplitude in the proposed framework using negative damping ratio is auto-adjustive and does not require to be pre-assumed, and eventually shows superiority on the aerodynamic damping identification and VIV prediction. Note that the proposed two-stage hybrid framework is also a potential alternative for flow-induced vibrations of many types of structures, such as the vibration of the cables (Chen et al., 2013, 2015; Gao et al., 2018; Jin et al., 2018), not limited to bridge deck discussed in this study.

Moreover, some limitations of the presented study should be noted. The vortex-induced force consists of both forced and self-excited components. The forced characteristic of VIV is ignored and only the self-excited component is considered in this study, and a more comprehensive model including both forced and self-excited vibration should be investigated in the future. The prediction fails to reproduce the secondary lock-in region observed in the wind tunnel tests, which suggests that the presented framework is not sensitive enough to capture the relatively weak aerodynamic self-excitation vibration and if the forced vibration characteristics of the deck is significant, obvious errors may occur. The presented prediction is also based on the linear superposition of aerodynamic damping and structural damping, which may not always be the actual situation. The 2-dimensional scheme of the divergent

simulation also suggests that the presented framework is not able to handle the effect caused by complex spatial turbulence structure in span-wise direction on bridge deck.

CRedit authorship contribution statement

Yubing Song: Writing – review & editing, Software, Writing – original draft, Formal analysis. **Zilong Ti:** Conceptualization, Validation, Writing – review & editing, Supervision. **Yongle Li:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The financial support from the National Natural Science Foundation of China (52008349), the Postdoctoral Science Foundation of China (2020M683356, 2021T140573), and the Natural Science Foundation of Sichuan Province (22NSFSC0857) are greatly appreciated by the authors.

References

- Bruno, L., Khris, S., 2003. The validity of 2D numerical simulations of vortical structures around a bridge deck. *Math. Comput. Model.* 37, 795–828.
- Brusiani, F., de Miranda, S., Patruno, L., Ubertini, F., Vaona, P., 2013. On the evaluation of bridge deck flutter derivatives using RANS turbulence models. *J. Wind Eng. Ind. Aerod.* 119, 39–47.
- Chen, W.-L., Xin, D.-B., Xu, F., Li, H., Ou, J.-P., Hu, H., 2013. Suppression of vortex-induced vibration of a circular cylinder using suction-based flow control. *J. Fluid Struct.* 42, 25–39.
- Chen, W.-L., Zhang, Q.-Q., Li, H., Hu, H., 2015. An experimental investigation on vortex induced vibration of a flexible inclined cable under a shear flow. *J. Fluid Struct.* 54, 297–311.
- Chen, X., 2013. Estimation of stochastic crosswind response of wind-excited tall buildings with nonlinear aerodynamic damping. *Eng. Struct.* 56, 766–778.
- Feng, C., 1968. The measurement of Vortex Induced Effects in Flow Past Stationary and Oscillating Circular and D-Section Cylinders. University of British Columbia.
- Ferziger, J.H., Perić, M., Street, R.L., 2002. *Computational Methods for Fluid Dynamics*. Springer.
- Gao, D., Chen, W., Eloy, C., Li, H., 2018. Multi-mode responses, rivulet dynamics, flow structures and mechanism of rain-wind induced vibrations of a flexible cable. *J. Fluid Struct.* 82, 154–172.
- Gao, G., Zhu, L., Li, J., Han, W., 2020. Application of a new empirical model of nonlinear self-excited force to torsional vortex-induced vibration and nonlinear flutter of bluff bridge sections. *J. Wind Eng. Ind. Aerod.* 205, 104313.
- Ge, Y., Zhao, L., Cao, J., 2022. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J. Wind Eng. Ind. Aerod.* 220, 104866.
- Ge, Y.J., Xiang, H.F., 2008. Computational models and methods for aerodynamic flutter of long-span bridges. *J. Wind Eng. Ind. Aerod.* 96, 1912–1924.
- Hallak, P.H., Pfeil, M.S., de Oliveira, S.R.C., Battista, R.C., de Sampaio, P.A.B., Bezerra, C. M.N., 2013. Aerodynamic behavior analysis of Rio-Niterói bridge by means of computational fluid dynamics. *Eng. Struct.* 56, 935–944.
- He, X., Kang, X., Yan, L., Flay, R.G., Ren, P., Wu, T., 2021. Numerical investigation of flow structures and aerodynamic interference around stationary parallel box girders. *J. Wind Eng. Ind. Aerod.* 215, 104610.
- Iwamoto, M., Fujino, Y., 1995. Identification of flutter derivatives of bridge deck from free vibration data. *J. Wind Eng. Ind. Aerod.* 54 (55), 55–63.
- Jeong, J., Hussain, F., 1995. On the identification of a vortex. *J. Fluid Mech.* 285, 69–94.
- Jin, X., Cheng, P., Chen, W.-L., Li, H., 2018. Prediction model of velocity field around circular cylinder over various Reynolds numbers by fusion convolutional neural networks based on pressure on the cylinder. *Phys. Fluids* 30.
- Laima, S., Jiang, C., Li, H., Chen, W., Ou, J., 2018a. A numerical investigation of Reynolds number sensitivity of flow characteristics around a twin-box girder. *J. Wind Eng. Ind. Aerod.* 172, 298–316.
- Laima, S., Li, H., 2015. Effects of gap width on flow motions around twin-box girders and vortex-induced vibrations. *J. Wind Eng. Ind. Aerod.* 139, 37–49.
- Laima, S., Li, H., Chen, W., Li, F., 2013. Investigation and control of vortex-induced vibration of twin box girders. *J. Fluid Struct.* 39, 205–221.
- Laima, S., Li, H., Chen, W., Ou, J., 2018b. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J. Fluid Struct.* 77, 115–133.
- Li, J., Wu, P., Hao, J., Pan, H., 2022. Experimental and numerical studies on the two “lock-in” regions characteristic of vertical vortex-induced vibration of H-shaped composite bridge deck. *J. Wind Eng. Ind. Aerod.* 228, 105122.
- Liu, S., Zhao, L., Fang, G., Hu, C., Ge, Y., 2021. Investigation on aerodynamic force nonlinear evolution for a central-slotted box girder under torsional vortex-induced vibration. *J. Fluid Struct.* 106, 103380.
- Nariman, N.A., 2016. Influence of fluid-structure interaction on vortex induced vibration and lock-in phenomena in long span bridges. *Front. Struct. Civ. Eng.* 10, 363–384.
- Naudascher, E., Wang, Y., 1993. Flow-induced vibrations of prismatic bodies and grids of prisms. *J. Fluid Struct.* 7, 341–373.
- Noguchi, K., Ito, Y., Yagi, T., 2020. Numerical evaluation of vortex-induced vibration amplitude of a box girder bridge using forced oscillation method. *J. Wind Eng. Ind. Aerod.* 196, 104029.
- Pan, Z.Y., Cui, W.C., Miao, Q.M., 2007. Numerical simulation of vortex-induced vibration of a circular cylinder at low mass-damping using RANS code. *J. Fluid Struct.* 23, 23–37.
- Sarwar, M.W., Ishihara, T., 2010. Numerical study on suppression of vortex-induced vibrations of box girder bridge section by aerodynamic countermeasures. *J. Wind Eng. Ind. Aerod.* 98, 701–711.
- Scanlan, R.H., 1978. The action of flexible bridges under wind, I: flutter theory. *J. Sound Vib.* 60, 187–199.
- Scanlan, R.H., Tomko, J.J., 1971. Airfoil and bridge deck flutter derivatives. *J. Eng. Mech. Div.* 97, 1717–1737.
- Shimada, K., Ishihara, T., 1999. Prediction of aeroelastic vibration of rectangular cylinders by k- ϵ model. *J. Aero. Eng.* 12, 122–135.
- Tamura, T., Miyagi, T., Kitagishi, T., 1998. Numerical prediction of unsteady pressures on a square cylinder with various corner shapes. *J. Wind Eng. Ind. Aerod.* 74, 531–542.
- Wang, Y., Chen, X., 2022. Extraction of aerodynamic damping and prediction of vortex-induced vibration of bridge deck using CFD simulation of forced vibration. *J. Wind Eng. Ind. Aerod.* 224, 104982.
- Wang, Y., Chen, X., Li, Y., 2020. Nonlinear self-excited forces and aerodynamic damping associated with vortex-induced vibration and flutter of long span bridges. *J. Wind Eng. Ind. Aerod.* 204, 104207.
- Wilcox, D.C., 1998. *Turbulence Modeling for CFD*. DCW industries La Canada, CA.
- Wu, T., Kareem, A., 2012. An overview of vortex-induced vibration (VIV) of bridge decks. *Front. Struct. Civ. Eng.* 6, 335–347.
- Xu, F., 2017. Free vibration numerical simulation technique for extracting flutter derivatives of bridge decks. *J. Wind Eng.* 12.
- Xu, K., Zhao, L., Ge, Y., 2017. Reduced-order modeling and calculation of vortex-induced vibration for large-span bridges. *J. Wind Eng. Ind. Aerod.* 167, 228–241.
- Yang, Y., Ma, T., Ge, Y., 2015. Evaluation on Bridge Dynamic Properties and VIV Performance Based on Wind Tunnel Test and Field Measurement, vol. 20. *Wind & Structures*, pp. 719–737.
- Zhang, T., Sun, Y., Li, M., Yang, X., 2020. Experimental and numerical studies on the vortex-induced vibration of two-box edge girder for cable-stayed bridges. *J. Wind Eng. Ind. Aerod.* 206, 104336.